CIFAR-10 IMAGE CLASSIFICATION

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import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.datasets import cifar10
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten, Dropout
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.optimizers import Adam, SGD

from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns

(X_train, y_train), (X_test, y_test) = cifar10.load_data()

print("Training Images:", X_train.shape)
print("Testing Images:", X_test.shape)

class_names = ['Airplane', 'Automobile', 'Bird', 'Cat', 'Deer',
               'Dog', 'Frog', 'Horse', 'Ship', 'Truck']

plt.figure(figsize=(10, 5))

for i in range(10):
    plt.subplot(2, 5, i + 1)
    plt.imshow(X_train[i])
    plt.title(class_names[y_train[i][0]])
    plt.axis("off")

plt.show()

X_train = X_train.astype('float32') / 255.0
X_test = X_test.astype('float32') / 255.0

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X_test = X_test.astype('float32') / 255.0

model = Sequential()

model.add(Flatten(input_shape=(32, 32, 3)))

model.add(Dense(1024, activation='relu'))
model.add(Dropout(0.3))

model.add(Dense(512, activation='relu'))
model.add(Dropout(0.3))

model.add(Dense(256, activation='relu'))
model.add(Dropout(0.3))

model.add(Dense(10, activation='softmax'))

model.compile(optimizer='adam',
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              loss='categorical_crossentropy',
              metrics=['accuracy'])

model.summary()
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